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selected works

Preprint 2026 — **AlphaNIST: Emergent Assembly Skills from Binary Task Rewards**
A binary success reward alone teaches one policy to assemble the full NIST board, with emergent regrasping and sequencing.

Preprint 2026 — **A Balanced Data Diet: Addressing the Bottleneck in Mega-Scale RL for Robot Control**
Balancing the data mix, not just scaling, enables effective learning with >1M+ parallel rollouts.

ICRA 2025 — **Using non-expert data to robustify imitation learning via offline reinforcement learning**
Offline RL on cheap non-expert data makes imitation policies robust to mistakes experts never demonstrate.

ICLR 2025 — **Model predictive adversarial imitation learning for planning from observation** · *Best Paper Finalist @ CoRL WS*
Pairing model-predictive control with adversarial imitation lets a robot plan from observation alone, no action labels.

CoRL 2025 — **Demonstrating Wheeled Lab: Modern Sim2Real for Low-cost, Open-source Wheeled Robotics**

ICRA 2024 — **Toward Self-Righting and Recovery in the Wild: Challenges and Benchmarks**

ICRA 2024 — **Open X-Embodiment: Robotic learning datasets and RT-X models** · *Best Paper Award*

RSS 2024 — **DROID: A large-scale in-the-wild robot manipulation dataset**

CoRL WORKSHOP 2024 — **Parental Guidance: Efficient Lifelong Learning through Evolutionary Distillation** · *Best Paper Award*

ICRA 2023 — **GuILD: Guided Incremental Local Densification for Accelerated Sampling-based Motion Planning**

IJRR 2018 — **Natural language instructions for human-robot collaborative manipulation**

Full publication list, videos, and writing at **rosarioscalise.com**.

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